Name:	Class:	Date given:
		Date due in:

Balancing **Equations**

1. Write the word equations as symbol equations. [3]

   a. hydrogen + oxygen ⟶ water

   b. magnesium + oxygen ⟶ magnesium oxide

   c. aluminium oxide ⟶ aluminium + oxygen

2. Rewrite the equations below so that they are balanced. [3]

   a. $H_2 + O_2 \longrightarrow H_2O$

   b. $Mg + O_2 \longrightarrow MgO$

   c. $Al_2O_3 \longrightarrow Al + O_2$

3. Write the word equations as balanced symbol equations. [4]

   a. sodium + hydrochloric acid ⟶ sodium chloride + hydrogen

   b. iron oxide + carbon ⟶ iron + carbon dioxide

   Fe_2O_3

Learning Outcomes (tick if achieved)

Q1	I can write symbol equations of chemical reactions	
Q2	I can balance equations of chemical reactions	
Q3	I can write balanced equations of chemical reactions	

Balancing Equations **Answers**

1. Write the word equations as symbol equations. [3]

   a. hydrogen + oxygen ⟶ water

   $H_2 + O_2 \longrightarrow H_2O$

   b. magnesium + oxygen ⟶ magnesium oxide

   $Mg + O_2 \longrightarrow MgO$

   c. aluminium oxide ⟶ aluminium + oxygen

   $Al_2O_3 \longrightarrow Al + O_2$ (students may write aluminium oxide as 'AlO')

2. Rewrite the equations below so that they are balanced. [3]

   a. $H_2 + O_2 \longrightarrow H_2O$

   2$H_2 + O_2 \longrightarrow$ **2**H_2O

   b. $Mg + O_2 \longrightarrow MgO$

   2$Mg + O_2 \longrightarrow$ **2**MgO

   c. $Al_2O_3 \longrightarrow Al + O_2$

   2$Al_2O_3 \longrightarrow$ **4**$Al +$ **3**O_2

3. Write the word equations as balanced symbol equations. [4]

   a. sodium + hydrochloric acid ⟶ sodium chloride + hydrogen

   $2Na + 2HCl > 2NaCl + H_2$

   b. iron oxide + carbon ⟶ iron + carbon dioxide

   2Fe_2O_3 **+ 3C ⟶ 4Fe + 3**CO_2